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Optimization of DualGAN for Image-to-Image Translation using Warm-Start Training

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In partial fulfillment of the requirement for the degree of Master in Information Technology (MIT)

Under the Supervision of

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STUDENT'S DECLARATION

I hereby declare that I am the sole author of this work, and that all sources of information and references used have been properly acknowledged and listed in this document. No other sources or assistance have been used beyond those cited.

_____

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SUPERVISOR'S RECOMMENDATION

I hereby recommend that this dissertation, titled "**Optimization of DualGAN for Image-to-Image Translation using Warm-Start Training**" prepared under my supervision by Mr. Avishek Kuinkel, in partial fulfillment of the requirements for the degree of Master in Information Technology (MIT), be submitted for evaluation.

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LETTER OF APPROVAL

We hereby certify that we have read this dissertation and, in our opinion, it meets the standards of scope and quality expected for a dissertation in partial fulfillment of the requirements for the degree of Master in Information Technology (MIT).

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PLAGIARISM TEST REPORT

ABSTRACT

This dissertation investigates a minimal-intervention strategy to enhance DualGAN for image-to-image (I2I) translation by introducing a short warm-start on limited paired data followed by standard unpaired DualGAN training, with no architectural or loss changes. The approach is motivated by practical settings where a small number of pairs co-exist with large unpaired collections (e.g., Facades labels↔photos and Day↔Night scenes). The method retains DualGAN's U-Net generators and PatchGAN discriminators and preserves optimizer state across the transition; only the data loader switches at a predefined epoch *k*. Experiments on Facades and Day-Night evaluate realism and fidelity using FID, SSIM, and MSE. Warm-start consistently improves distributional realism: FID improves from 282.46 → 190.96 on Facades and 421.12 → 386.89 on Day-Night. On Day-Night, warm-start also yields higher structural similarity (SSIM 0.0793 → 0.2121) with a small MSE trade-off; on Facades, the baseline attains lower MSE/SSIM but appears less realistic distributionally. Loss curves show a stable transition at epoch *k* without spikes, indicating that a brief supervised anchor accelerates early alignment while preserving DualGAN's unsupervised learning benefits. Overall, warm-start is a simple, compute-neutral enhancement that improves perceptual quality when pairs are scarce and can be tuned via *k* and the cycle-loss weight λ.

Keywords: Image-to-Image Translation, DualGAN, Warm-Start Mechanism

TABLE OF CONTENTS

ACKNOWLEDGEMENT iv

PLAGIARISM TEST REPORT v

ABSTRACT vi

LIST OF TABLES x

LIST OF FIGURES xi

LIST OF ABBREVIATIONS xii

UNIT 1: INTRODUCTION 1

1.1 Introduction 1

1.2 Problem Statement 1

1.3 Objectives 2

1.4 Scope and Limitation 2

1.5 Report Organization 3

UNIT 2: BACKGROUND STUDY AND LITERATURE REVIEW 4

2.1 Background Study 4

2.2 Literature Review 4

UNIT 3: METHODOLOGY 6

3.1 Description of Methodology 6

3.2 Data Set Description 8

3.3 Data Preprocessing 8

3.4 Description of DualGAN Algorithms 9

3.4.1 Network Architecture 9

3.4.2 Two-Phase Training Procedure 10

3.4.3 Pseudocode of DualGAN with Cycle Consistency) 11

3.4.4 Mathematical Formulation 12

3.4.5 Hyperparameter Tuning 12

3.5 Performance/Evaluation Metrics 13
3.5.1 Mean Square Error (MSE) 13
3.5.2 Structural Similarity Index (SSIM) 14
3.5.3 Frechet Inception Distance (FID) 14
UNIT 4: IMPLEMENTATION 16
4.1 Tools and Techniques 16
4.2 Implementation Details 17
4.2.1 Python Environment and Dependencies 17
4.2.2 Model Architecture and Code Structure 17
4.2.3 Hyperparameter Configuration 19
4.2.4 Two-Phase Training Implementation 20
4.2.5 Python Implementation of Evaluation Metrics 21
UNIT 5: RESULT AND ANALYSIS 23
5.1 Findings and Observations 23
5.1.1 Quantitative results (primary tables) 23
5.1.2 Training dynamics (epoch analysis) 23
5.1.3 Qualitative observations (visual evidence) 25
5.2 Result Analysis 28
5.2.1 Comparative effectiveness 28
5.2.2 Why the trade-off appears (mechanistic view) 28
5.2.3 Sensitivity to schedule and weights (evidence-based guidance) 28
5.2.4 Error analysis and remedies 29
5.2.5 Efficiency and stability 29
5.2.6 Overall conclusion 29
UNIT 6: CONCLUSION AND FUTURE RECOMMENDATIONS 30
6.1 Conclusion 30

6.2 Future Recommendations 30

REFERENCES 32

LIST OF TABLES

Table 4.1 Summary of tools used 16

Table 4.2 Hyperparameter Setup for Day-Night and Facades Datasets 19

Table 5.1 Facades - test metrics (n = 100 images, mean) 23

Table 5.2 Day-Night - validation metrics (n = 10 pairs, mean). 23

LIST OF FIGURES

Figure 3.1 Block Diagram of Research Methodology 6

Figure 3.2 Network Architecture and data flow chart of Dual-GAN for I2I translation 9

Figure 5.1 Facades - Warm-Start training losses. 24

Figure 5.2 Facades - Baseline training losses 24

Figure 5.3 Day-Night - Warm-Start training losses. 24

Figure 5.4 Day-Night - Baseline training losses. 25

Figure 5.5 Qualitative translation results on Facades (Warm Start). 25

Figure 5.6 Qualitative translation results on Facades (Baseline). 26

Figure 5.7 Qualitative translation results on Day-Night (Warm Start). 27

Figure 5.8 Qualitative translation results on Day-Night (Baseline). 27

LIST OF ABBREVIATIONS

cGAN	Conditional GAN
DualGAN	Dual Generative Adversarial Network
FID	Fréchet Inception Distance
GAN	Generative Adversarial Network
I2I	Image-to-Image
L1	L1 Reconstruction Loss
MSE	Mean Squared Error
PatchGAN	Patch-based Discriminator
SSIM	Structural Similarity Index
U-Net	U-Shaped Network
WGAN	Wasserstein GAN

UNIT 1: INTRODUCTION

1.1 Introduction

Image-to-image (I2I) translation has become a fundamental task in computer vision, with applications in style transfer, object transfiguration, and semantic label manipulation [1], [2]. Among unsupervised approaches, Dual Generative Adversarial Networks (GAN) enables learning of bidirectional mappings between two domains using only unpaired data by combining adversarial training with cycle-consistency constraints [1].

While DualGAN eliminates the need for paired supervision during the main training process, its convergence is relatively slow, and the quality of generated images often suffers from geometric distortions and loss of fine details, especially on structured domains such as architectural facades and day-to-night scene translation [1], [3]. In many real-world scenarios, however, a modest number of paired examples are readily available (e.g., the 606 registered label ↔ photo pairs in the Facades dataset [4] or temporally aligned day ↔ night frames from driving videos [5]).

This observation motivates a simple yet underexplored question whether or not the original DualGAN architecture benefit from a short initial training phase on paired data before continuing with its standard unsupervised protocol on large unpaired collections.

This thesis proposes and systematically evaluates a minimal-intervention, two-phase training strategy termed “warm-start training” for DualGAN. The method uses a few epochs of paired supervision (employing the exact same adversarial and cycle losses as the original DualGAN) followed by uninterrupted unsupervised training without any architectural modification, additional loss terms, or discriminator re-purposing.

1.2 Problem Statement

Training DualGAN from scratch on purely unpaired data results in slow convergence and sub-optimal translation quality on domains requiring precise structural preservation, such as label ↔ facade and day ↔ night translation [1]. Although fully supervised frameworks like pix2pix achieve superior fidelity [2], collecting large paired datasets remains costly and impractical in many cases.

While semi-supervised and weak-supervision extensions of GANs have been explored [6], [7], [8], most introduce architectural changes, pseudo-labeling mechanisms, or auxiliary

classification tasks. In contrast, this research investigates whether significant performance gains can be obtained through a far simpler approach: leveraging existing paired examples only for a brief warm-start phase (typically 5–10 epochs) and then resuming the original unsupervised DualGAN training schedule without any further modification.

1.3 Objectives

The primary objective of this thesis is:

- To optimize the performance of DualGAN for image-to-image translation by introducing a short warm-start pre-training phase using a limited amount of paired data.
- To seamlessly continue standard unsupervised training on unpaired data without any architectural or loss-function modifications.

1.4 Scope and Limitation

Scope

This research is confined to the original DualGAN architecture and loss functions with no architectural or objective modifications [1]. Experiments are conducted on only two datasets that naturally provide both paired and unpaired images:

- Facades dataset (architectural labels ↔ real photos) [4]
- Day-Night dataset (daytime scenes ↔ nighttime scenes) compiled from transient-attribute benchmarks and temporally aligned video frames [5]

The proposed warm-start training strategy is evaluated using the following quantitative metrics: Mean Squared Error (MSE), Fréchet Inception Distance (FID), and Structural Similarity Index (SSIM). Qualitative visual inspection is also performed. The warm-start phase duration is varied between 5 and 15 epochs using the official paired splits or their subsets.

Limitations

- The method requires a limited amount of paired data for the warm-start phase and is not a fully unsupervised approach.
- Evaluation is restricted to the Facades and Day-Night datasets only; generalization to other domains (e.g., natural photo-to-photo translation) is not studied.

- No comparisons are made with state-of-the-art methods developed after 2019 (e.g., contrastive-learning-based approaches).
- Computational resources are limited to Google Colab with a single NVIDIA T4 GPU, constraining the scale of hyperparameter search.

1.5 Report Organization

The dissertation is structured into the following six chapters:

Unit 1: Introduction Present the background of image-to-image translation, limitations of existing unsupervised methods, motivation for warm-start training in DualGAN, problem statement, objectives, scope, limitations, and organization of the report.

Unit 2: Literature Review Surveys the evolution from paired translation models (pix2pix) to unsupervised frameworks (CycleGAN, DualGAN, DistanceGAN), and recent contrastive and few-shot approaches, highlighting the research gap addressed by this work.

Unit 3: Methodology Describes the original DualGAN architecture, loss functions, the proposed two-phase warm-start training strategy, datasets (Day↔Night and CMP Facades), evaluation metrics (MSE, SSIM, FID), and experimental protocol.

Unit 4: Implementation Details the practical implementation in Google Colab using TensorFlow 2.19.0 and NVIDIA T4 GPU, data pipelines, model building with Keras, hyperparameter settings, training procedure for both warm-start and baseline models, and reproducibility measures.

Unit 5: Results and Analysis Presents quantitative results, training loss curves, qualitative translations, metric distributions, and critical discussion explaining the contrasting performance of warm-start on Day↔Night (successful) versus Labels↔Photos (counterproductive).

Unit 6: Conclusion and Future Recommendations Summarizes key findings, states the partial achievement of objectives, highlights the dataset-dependent nature of warm-start effectiveness, and suggests future directions including adaptive scheduling, paired-data augmentation, and integration with modern contrastive frameworks.

UNIT 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

Image-to-image (I2I) translation has evolved into a cornerstone task in computer vision, enabling applications such as style transfer, object transfiguration, and semantic label manipulation [1], [2]. Early approaches relied heavily on supervised learning with large paired datasets, which are expensive and often impractical to acquire in domains like autonomous driving or architectural imaging.

The introduction of unsupervised I2I translation methods marked a significant breakthrough by removing the requirement for paired training data. DualGAN [1] and Cycle GAN [2] pioneered this paradigm using cycle-consistency constraints to enforce structural correspondence without direct supervision. Despite their success, these models still exhibit slow convergence and limited fidelity on structured tasks (e.g., label ↔ photo and day ↔ night translation), where precise geometric and semantic preservation is critical [1], [3].

In many practical settings, a modest number of paired examples coexist with abundant unpaired data. This observation motivates the core idea of the present work: a minimal-intervention warm-start strategy that briefly exposes the original DualGAN to paired supervision before continuing standard unsupervised training, requiring no architectural or loss modifications.

2.2 Literature Review

GANs form the foundation of modern generative modelling [8]. A generator and a discriminator are trained in an adversarial manner: the **generator** learns to produce realistic samples, while the **discriminator** distinguishes real data from generated samples. This minimax framework has revolutionized numerous computer vision tasks, including image synthesis and translation [9].

Conditional GANs (cGANs), particularly pix2pix [2], extended GANs to supervised I2I translation by conditioning both the generator and discriminator on input images. The pix2pix framework employs a U-Net generator and a PatchGAN discriminator, combined with an L1 loss to enforce pixel-wise correspondence. While highly effective, pix2pix requires large paired datasets, limiting its applicability in data-scarce scenarios [2].

To address training instability in original GANs, Wasserstein GAN (WGAN) [3] replaced Jensen-Shannon divergence with the Earth Mover distance, enforced via weight clipping or gradient penalty. This modification significantly improves convergence and reduces mode collapse, influencing subsequent unsupervised translation models [3].

DualGAN is a pioneering unsupervised I2I translation framework that operates without any paired data. It consists of two symmetric GANs: one translating from domain U→V and the other from V→U. By combining Wasserstein adversarial loss with cycle-consistency loss, DualGAN establishes a closed translation loop that enforces bidirectional reconstruction. Its key advantage lies in its ability to train solely on unpaired collections while achieving competitive results on various tasks [1].

Several works have explored semi-supervised and weakly supervised extensions of I2I translation. The Author [10] introduced semantic consistency losses and segmentation mask guidance to preserve object structure across domains. The research [6] proposed a two-pass approach that generates pseudo-labels via unsupervised translation and refines them with limited supervision. The paper [7] presented SUIT, an unsupervised adaptation method that enforces semantic consistency through a dedicated content loss. The research paper [8] extended GAN discriminators to perform simultaneous classification in semi-supervised settings.

While these methods achieve strong performance by introducing new losses, auxiliary networks, or pseudo-labeling mechanisms, they increase model complexity. In contrast, the present work investigates a far simpler alternative: using a limited number of paired examples only for a short initial warm-start phase (3–15 epochs) on the original DualGAN, then resuming standard unsupervised training with no further changes. This minimal-intervention strategy has received little attention in the literature despite its practical appeal.

UNIT 3: METHODOLOGY

3.1 Description of Methodology

This research proposes a semi-supervised enhancement of DualGAN by introducing a two-phase training strategy that use both limited paired data and larger unpaired collections. The core idea is to **warm-start** the DualGAN model using paired examples and then continue training in an unsupervised manner on unpaired data, while keeping the underlying architecture and objective functions unchanged.

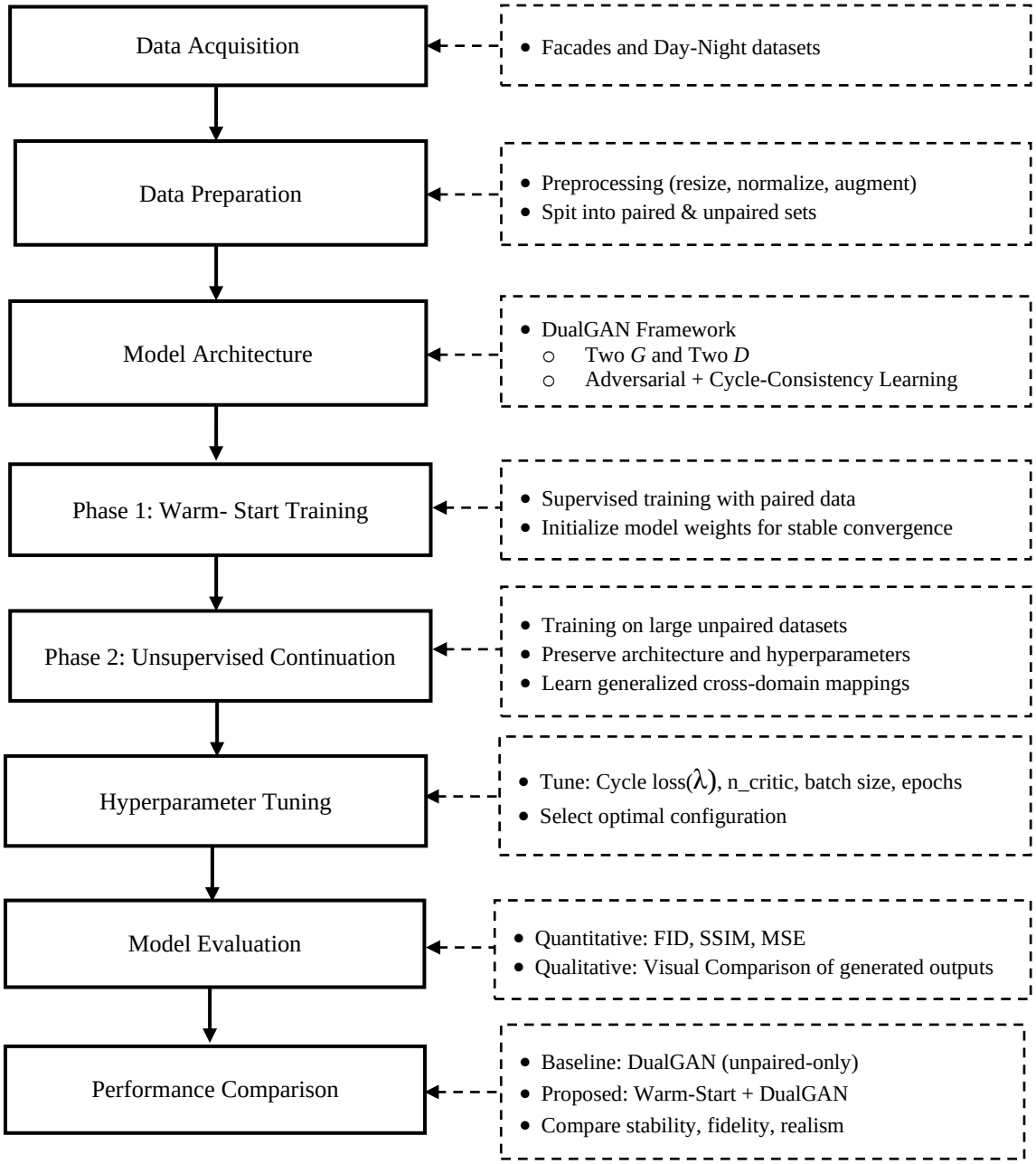


Figure 3.1 Block Diagram of Research Methodology

Figure 3.1 presents the overall research methodology. The main stages are:

- **Data Acquisition**: Images are collected from benchmark semantic image-to-image (I2I) datasets covering two translation tasks: facades, and day-night scenes.
- **Data Preparation**: All datasets are preprocessed to a consistent resolution, normalized pixel range, and standard partitioning into training/validation/test sets. For training, the data are also organized into **paired** subsets (for warm-start) and **unpaired** subsets (for fully unsupervised DualGAN learning).
- **Model Architecture (DualGAN)**: A DualGAN framework is adopted with two generators (A→B, B→A) and two discriminators (for Domains A and B). The model is trained using adversarial and cycle-consistency objectives to learn bidirectional mappings between domains.
- **Phase 1 – Warm-Start Training (Supervised):** In the first phase, the DualGAN model is trained on paired examples (A, B). This supervised signal provides an initial alignment between domains and improves stability of subsequent unsupervised training.
- **Phase 2 – Unsupervised Continuation (DualGAN):** After the warm-start, training continues on larger **unpaired** datasets from each domain using the standard DualGAN objective. Network weights, optimizer state, and hyperparameters are preserved; only the data loader is switched from paired to unpaired samples.
- **Hyperparameter Tuning**: Key hyperparameters (e.g., batch size, n_critic, cycle-loss weight λ, transition epoch k, effective data size, dropout rate) are tuned using validation performance to obtain an empirically optimal configuration of the semi-supervised DualGAN.
- **Model Evaluation**: The trained models are evaluated using Mean Squared Error (MSE), Structural Similarity Index (SSIM), and Fréchet Inception Distance (FID), complemented by qualitative visual inspection of generated translations.
- **Performance Comparison and Effectiveness**: Finally, the proposed **Warm-Start DualGAN** is compared against a baseline **DualGAN** that is trained purely on unpaired data. The comparison quantifies the benefits of the warm-start mechanism and supports conclusions about model effectiveness for semantic I2I translation.

3.2 Data Set Description

Two publicly available paired datasets are used:

- **Facades [4]:** This dataset contains images of facades along with corresponding labels. The labels indicate various aspects of the facades, such as architectural style, materials, or presence of specific elements like windows or doors. The total number of training images is 400 and 100 test images. Each image is an RGB image with dimensions of 256 × 256.
- **Day-night [5]:** This dataset consists of image pairs captured of the same scene under different lighting conditions, typically day and night. The total number of training images is 80 and 10 test images. Each image is an RGB image with dimensions of 512 × 512.

Both datasets are resized to 256×256 during preprocessing for consistent training.

3.3 Data Preprocessing

Data preprocessing aims to produce consistent inputs across datasets and to support both the supervised warm-start and the unsupervised DualGAN training stages.

The main preprocessing steps are:

- **Resizing:** All images are resized to a common square resolution (typically 256×256 pixels). For day–night, the original 512×512 images are downscaled to this target resolution.
- **Normalization:** Pixel values are scaled to the range [−1, 1] by dividing by 127.5 and subtracting 1. This normalization stabilizes gradient magnitudes and accelerates convergence in deep networks.
- **Horizontal Flipping (Data Augmentation):** During training (but not testing), images are randomly flipped horizontally with probability 0.5. This simple augmentation increases effective data diversity and helps the model generalize across mirrored viewpoints.
- **Partitioning into Paired and Unpaired Sets:** For each dataset, images are partitioned into training, validation, and test splits. Within the training split, samples are further organized into:

- **Paired sets**, where exact correspondences *(a, b)* are preserved for use in the warm-start phase.
- **Unpaired sets**, where domain A and domain B images are drawn independently to simulate real-world unsupervised translation scenarios.

These preprocessing steps ensure that the DualGAN model receives uniform, well-scaled inputs across all three tasks, while preserving the semantic information needed for accurate cross-domain translation.

3.4 Description of DualGAN Algorithms

The DualGAN is the core model used to perform bidirectional image-to-image translation between two domains, A and B. It is particularly suitable for settings with limited paired data and larger unpaired collections.

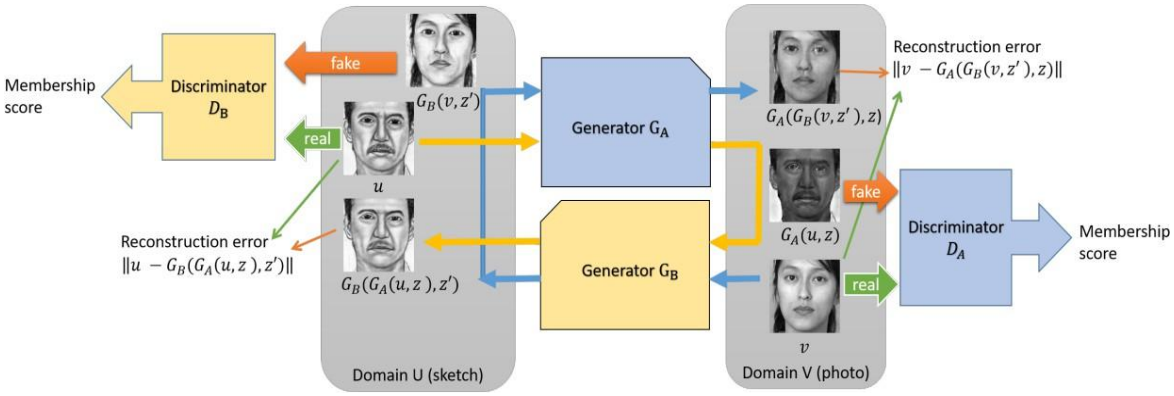


Figure 3.2 Network Architecture and data flow chart of Dual-GAN for I2I translation [1]

3.4.1 Network Architecture

DualGAN consists of **two generators** and **two discriminators**, one pair for each direction:

- **Generators**
  - G_{AB}: maps images from Domain A to Domain B (A→B).
  - G_{BA}: maps images from Domain B to Domain A (B→A).
- **Discriminators**
  - D_A: distinguishes real images in Domain A from fake images produced by G_{BA} .
  - D_B: distinguishes real images in Domain B from fake images produced by G_{AB}.

The two generators form a **dual (cycle) structure**: an image translated A→B→A should reconstruct the original A, and similarly B→A→B reconstructs B. This is enforced through a cycle-consistency (reconstruction) loss. In practice, the generators are implemented as U-Net-style encoder–decoder networks with skip connections, and the discriminators use a PatchGAN architecture that classifies local image patches as real or fake.

3.4.2 Two-Phase Training Procedure

To align with the methodology in **Figure 3.1**, DualGAN is trained using a hybrid supervised–unsupervised strategy comprising three conceptual stages: a supervised warm-start, a transition step, and an unsupervised continuation phase.

1. **Supervised warm-start Phase**
    - The model is first trained with **paired samples** *(a, b)* where a ∈ A and b ∈ B.
    - Generators G_{AB} and G_{BA} earn direct mappings between domains using exact correspondences.
    - Discriminators D_A and D_B learn to distinguish real images from those produced by the generators in their respective domains.
    - This phase is limited to a small number of epochs (e.g., $k \in \{3, 5, 10, 15\}$) and serves as a **warm-start**, providing a good initialization for later unsupervised learning.
2. **Transition Step**
    - Training is monitored using an iteration counter or epoch index.
    - Once the predefined transition epoch *k* is reached, the data loader is switched from paired training samples to unpaired samples.
    - Importantly, the **architecture, network weights, optimizer state, and hyperparameters** are kept unchanged. The model therefore continues learning from a well-aligned starting point.
3. **Unsupervised Phase**
    - In the second phase, the model is trained on unpaired samples from Domains A and B.
    - Generators G_{AB} and G_{BA} adapt to translate images without explicit one-to-one correspondences.

- Discriminators D_A and D_B refine their decision boundaries based on a wider variety of inputs.
- Training now relies purely on adversarial and cycle-consistency losses, as in the original DualGAN framework.

After both phases, the resulting DualGAN model integrates information learned from paired and unpaired data and can translate images between Domains A and B in both directions, even when direct pairings are not available at test time.

3.4.3 Pseudocode of DualGAN with Cycle Consistency)

A simplified training loop for DualGAN can be expressed as:

```
Initialize generators G_AB, G_BA and discriminators D_A, D_B

while not converged:
    # Sample mini-batch of images from each domain
    a ← sample_batch(A)          # images from Domain A
    b ← sample_batch(B)          # images from Domain B

    # Forward translation
    b_fake = G_AB(a)
    a_fake = G_BA(b)

    # Cycle reconstruction
    a_cycle = G_BA(b_fake)
    b_cycle = G_AB(a_fake)

    # Update discriminators D_A, D_B
    #   using real (a, b) and fake (a_fake, b_fake) samples

    # Update generators G_AB, G_BA
    #   using adversarial loss + dual (cycle) reconstruction loss

end while
```

In the **semi-supervised** setting used in this study, the early iterations additionally exploit paired *(a, b)* examples to compute supervised reconstruction losses (warm-start phase), before switching to purely unpaired sampling in the continuation phase.

3.4.4 Mathematical Formulation

Let $a \sim p_A(a)$ denote samples from Domain A and $b \sim p_B(b)$ denote samples from Domain B.

- **Adversarial losses**

Each generator tries to fool its corresponding discriminator. The adversarial losses for the two directions are:

$$\mathcal{L}_{GAN}(G_{AB}, D_B) = \mathbb{E}_{b\sim PB}(b)[\log D_B(b)] + \mathbb{E}_{a\sim PA}(a)\left[\log\left(1 - D_B\left(G_{AB}(a)\right)\right)\right]$$ ------- eq (1)

$$\mathcal{L}_{GAN}(G_{BA}, D_A) = \mathbb{E}_{a\sim PA}(a)[\log D_A(a)] + \mathbb{E}_{b\sim PB}(b)\left[\log\left(1 - D_A\left(G_{BA}(b)\right)\right)\right]$$ ------- eq (2)

These terms encourage G_{AB} and G_{BA} to produce outputs that are indistinguishable from real images in Domains B and A, respectively.

- **Dual (cycle) reconstruction loss**

To enforce consistency in both directions, DualGAN uses a dual (cycle) loss:

$$\mathcal{L}_{dual} = \mathbb{E}_{a\sim PA}(a)\left\|G_{BA}\left(G_{AB}(a)\right) - a\right\|_1 + \mathbb{E}_{b\sim PB}(b)\left\|G_{AB}\left(G_{BA}(b)\right) - b\right\|_1$$ ---------- eq (3)

This loss ensures that an image translated from one domain to the other and back again remains close to the original (i.e., $a \rightarrow b \rightarrow a$, $b \rightarrow a \rightarrow b$).

- **Total objective**

The overall objective combines both adversarial losses and the dual reconstruction loss:

$$\mathcal{L} = \mathcal{L}_{GAN}(G_{AB}, D_B) + \mathcal{L}_{GAN}(G_{BA}, D_A) + \lambda\,\mathcal{L}_{dual}$$ ---------- eq (4)

where $\lambda > 0$ is a weighting factor that balances **translation realism** (adversarial terms) against **cycle consistency** (dual loss). During training, the generators G_{AB}, G_{BA} and discriminators D_A, D_B are updated to minimize or maximize this objective as appropriate (generators minimize L, discriminators maximize their respective adversarial terms).

3.4.5 Hyperparameter Tuning

In line with the "Hyperparameter Tuning" block in **Figure 3.1**, several configuration parameters are tuned empirically:

- batch size and number of critic steps (n_{critic});
- learning rate and optimizer settings;
- cycle-loss weight λ (and, if applicable, separate λ_A and λ_B);

- warm-start transition epoch *k*;
- effective data size and dropout rate.

Candidate configurations are evaluated on a validation subset using MSE, SSIM, and FID, and the best-performing setting is carried forward to final training and testing.

3.5 Performance/Evaluation Metrics

In the context of semantic-based image-to-image translation, three metrics are employed to assess model performance, corresponding to the "Model Evaluation" block of **Figure 3.1**:

- **Mean Squared Error (MSE)**: measures pixel-wise reconstruction error between translated images and their ground-truth targets.
- **Structural Similarity Index (SSIM)**: measures perceptual similarity by comparing luminance, contrast, and structural information.
- **Fréchet Inception Distance (FID)**: assesses the realism and distributional alignment of generated images relative to real images in a deep feature space.

MSE provides a straightforward quantitative measure of error magnitude, SSIM captures perceptual fidelity, and FID evaluates both quality and diversity of generated images. Together, these metrics are used both for **hyperparameter selection** and for **comparative evaluation** between the proposed Warm-Start DualGAN and the baseline vanilla DualGAN.

3.5.1 Mean Square Error (MSE)

MSE calculates the average squared difference between the original and generated images. For two images *I* and *K* of size $m \times n$, it is defined as [11]:

$$MSE = \frac{1}{mn}\sum_{i=0}^{m-1}\sum_{j=0}^{n-1}[I(i,j) - K(i,j)]^2 \quad \text{---------- eq (5)}$$

Where:

- *m, n:* Dimensions of the images (width and height).
- *I(i, j):* Pixel value at position *(i, j)* in the original image.
- *K(i, j):* Pixel value at position *(i, j)* in the GAN-generated image

MSE is always non-negative; a value of 0 indicates a perfect match. Higher MSE values imply greater dissimilarity. In this study, we report both the **mean** and **median** of MSE values over the test set to summarize error behaviour.

3.5.2 Structural Similarity Index (SSIM)

SSIM evaluates perceived image quality by comparing local patterns of pixel intensities that have been normalized for luminance and contrast [11]. It decomposes similarity into three components:

Luminance, Contrast, and Structure Comparison Functions:

- **Luminance (*l(x,y)*):** Similarity in average brightness.
- **Contrast (*c(x,y)*):** Similarity in intensity variance.
- **Structure (*s(x,y)*):** Similarity in the structural patterns.

$$SSIM(x,y) = [l(x,y)]^{\alpha} \times [c(x,y)]^{\beta} \times [s(x,y)]^{\gamma}$$ ---------- eq (6)

Where:

- x and y are the two images being compared.
- α, β, and γ are parameters to adjust the relative importance of each component.

In this research, both **Average SSIM (ASSIM)** and **Median SSIM (MSSIM)** are computed over the test set. ASSIM summarizes overall similarity, while MSSIM is less sensitive to outliers in local patches.

3.5.3 Frechet Inception Distance (FID)

The Fréchet Inception Distance (FID) is a widely used metric for evaluating GAN-based image synthesis and translation [11]. It compares the distribution of generated images with that of real images in a deep feature space, typically taken from an intermediate layer of the Inception network.

Let (μr, Σr) and (μg, Σg) denote the mean and covariance of Inception feature vectors for real and generated images, respectively. The FID score is defined as:

$$FID = \|\mu r - \mu g\|^2 + Tr(\sum r + \sum g - 2\sqrt{(\sum r \sum g)})$$ ---------- eq (7)

Where:

- $\|\mu r - \mu g\|^2$ is the squared Euclidean distance between the mean vectors of real and generated images.
- Tr denotes the trace of the matrix.
- Σr and Σg are the covariance matrices of the real and generated images, respectively.

- The term $2\sqrt{(\sum r \sum g)}$ involves the square root of the product of the two covariance matrices, which is then multiplied by 2.

In this study, FID complements MSE and SSIM by capturing both the **realism** and the **diversity** of generated images. It is used to compare the proposed Warm-Start DualGAN against the DualGAN baseline across all two translation tasks.

UNIT 4: IMPLEMENTATION

4.1 Tools and Techniques

Warm-Start DualGAN was implemented in **Google Colab Pro** to leverage managed GPU acceleration and a reproducible, shareable runtime.

Hardware Specifications:

- **GPU Accelerator**: NVIDIA T4 with 15 GB VRAM (enabled for TensorFlow GPU operations)
- **System RAM**: 12.7 GB
- **Storage**: 235 GB temporary SSD (for datasets, caches, and checkpointing)

Software and Libraries:

- **Language:** Python **3.12**
- **DL framework: TensorFlow 2.19.0 (GPU)** for model definition, training loops, and mixed-precision when applicable
- **Data utilities:** NumPy (tensor ops), Pandas (metadata / splits), OpenCV (image I/O & resize)
- **Monitoring & visuals:** Matplotlib (loss plots, grids), TensorBoard (scalars, images, checkpoints)
- **Optimization:** Adam (β_1=0.5, β_2=0.999), optional gradient-penalty when using a Wasserstein objective

Colab runtimes were configured with CUDA-compatible TensorFlow wheels and pip-installed dependencies directly in the notebook.

Table 4.1 Summary of tools used

Tool / Library	**Role in implementation**
Python 3.12	Orchestration, data pipelines, experiments
TensorFlow 2.19.0 (GPU)	DualGAN architecture, losses, training/evaluation loops
NumPy / Pandas	Array ops, dataset manifests, splits

Tool / Library	Role in implementation
OpenCV	Image I/O, resize to target resolution, basic augmentation
Matplotlib / Seaborn	Loss curves, qualitative grids
TensorBoard	Live metrics, images, checkpoints, HP tuning traces

These tools enabled consistent preprocessing across datasets and efficient GPU utilization during both warm-start and unsupervised stages.

4.2 Implementation Details

The entire research was conducted using Python 3.12 in Google Colab Pro environment with an NVIDIA T4 GPU (16 GB VRAM). TensorFlow 2.19.0 with Keras high-level API was chosen as the primary deep-learning framework due to its robust GPU support, mature mixed-precision capabilities, and flexibility for custom training loops.

4.2.1 Python Environment and Dependencies

The following core Python libraries were utilised:

```
import tensorflow as tf
import numpy as np, cv2, glob, os, pandas as pd
import matplotlib.pyplot as plt
from skimage.metrics import structural_similarity as ssim
import subprocess, shutil
from tqdm import tqdm
```

4.2.2 Model Architecture and Code Structure

The original DualGAN architecture [1] was implemented without modification:

- **Generators (G_AB, G_BA):** U-Net with skip connections (encoder–decoder structure, 8 down-sampling / upsampling blocks)

```
def build_generator():
    inputs = tf.keras.layers.Input(shape=[256, 256, 3])
    down_stack = [downsample(64, 4, apply_batchnorm=False),
                  downsample(128, 4), downsample(256, 4),
                  downsample(512, 4), downsample(512, 4),
                  downsample(512, 4), downsample(512, 4),
```

```
                  downsample(512, 4)]  # bottleneck (1×1×512)
    up_stack = [upsample(512, 4, apply_dropout=True),
                upsample(512, 4, apply_dropout=True),
                upsample(512, 4, apply_dropout=True),
                upsample(512, 4), upsample(256, 4),
                upsample(128, 4), upsample(64, 4)]
    last = tf.keras.layers.Conv2DTranspose(3, 4, strides=2,
                                           padding='same',
activation='tanh')
    x = inputs; skips = []
    for down in down_stack:
        x = down(x); skips.append(x)
    skips = reversed(skips[:-1])
    for up, skip in zip(up_stack, skips):
        x = up(x); x = tf.keras.layers.Concatenate()([x, skip])
    return tf.keras.Model(inputs=inputs, outputs=last(x))
```

- **Discriminators (D_A, D_B):** PatchGAN (70×70 receptive field, 4 convolutional layers)

```
def build_discriminator():
    model = tf.keras.Sequential([
        tf.keras.layers.Conv2D(64, 4, strides=2, padding='same',
                               input_shape=[256,256,3]),
        tf.keras.layers.LeakyReLU(),
        tf.keras.layers.Conv2D(128, 4, strides=2, padding='same'),
        tf.keras.layers.BatchNormalization(),
        tf.keras.layers.LeakyReLU(),
        tf.keras.layers.Conv2D(256, 4, strides=2, padding='same'),
        tf.keras.layers.BatchNormalization(),
        tf.keras.layers.LeakyReLU(),
        tf.keras.layers.Conv2D(512, 4, strides=1, padding='same'),
        tf.keras.layers.BatchNormalization(),
        tf.keras.layers.LeakyReLU(),
        tf.keras.layers.Conv2D(1, 4, strides=1, padding='same')
    ])
    return model
```

Models were defined using the **Keras** functional API and compiled within a custom training loop using **tf.GradientTape**.

4.2.3 Hyperparameter Configuration

The same hyperparameter set was used for both datasets and across both training phases (warm-start and unsupervised continuation) to ensure fair comparison. These values were adopted from the original DualGAN paper [1] and fine-tuned empirically on validation data. **Table 4.2** summarizes the key parameters, which remained unchanged across both phases and datasets.

Table 4.2 Hyperparameter Setup for Day-Night and Facades Datasets

Hyperparameter	**Value**	**Remarks**
Image resolution	256 × 256 × 3	All images resized to this size
Batch size	1	Limited by T4 GPU memory at 256×256
Optimizer	Adam (β_1=0.5, β_2=0.999)	As in original DualGAN paper
Learning rate (G & D)	0.0002	Linear decay after 100 epochs (not reached)
Cycle consistency weight (λ)	10	Controls reconstruction vs. realism trade-off
Discriminator updates (n_critic)	5	Wasserstein GAN with gradient penalty
Gradient penalty coefficient	10	Standard WGAN-GP setting
Warm-start epochs (Facades)	10	On 36 paired images
Warm-start epochs (Day–Night)	5	On 90 paired images
Total epochs (Facades)	30	10 warm start + 20 unsupervised

Hyperparameter	Value	Remarks
Total epochs (Day–Night)	~60	5 warm start + continuation on 80 unpaired images

4.2.4 Two-Phase Training Implementation

The warm-start mechanism was implemented via a simple conditional switch inside the training loop, preserving optimizer momentum across phases:

Code Snippet - Warm-start phase switch

```
warm_start_epochs = 10 if "facades" in dataset_name else 5

for epoch in range(1, total_epochs + 1):
    print(f"\n[{'Warm' if epoch <= warm_start_epochs else 'Full'}] Epoch
{epoch}")

    if epoch <= warm_start_epochs:
        # Phase 1: Paired supervision
        for real_a, real_b in paired_dataset.take(steps_per_epoch):
            train_step(real_a, real_b, is_paired=True)
    else:
        # Phase 2: Standard unsupervised DualGAN
        for real_a in domain_a_dataset.take(steps_per_epoch//2):
            for real_b in domain_b_dataset.take(steps_per_epoch//2):
                train_step(real_a, real_b, is_paired=False)
```

Code Snippet – Core train_step() with Wasserstein + Cycle loss

```
@tf.function
def train_step(real_a, real_b, is_paired=True):
    with tf.GradientTape(persistent=True) as tape:
        fake_b = gen_AB(real_a, training=True)
        fake_a = gen_BA(real_b, training=True)
        cycled_a = gen_BA(fake_b, training=True)
        cycled_b = gen_AB(fake_a, training=True)

        disc_loss_A = discriminator_loss(disc_A, real_a, fake_a)
        disc_loss_B = discriminator_loss(disc_B, real_b, fake_b)
```

```
        gen_loss_AB = generator_loss(disc_B, fake_b)
        gen_loss_BA = generator_loss(disc_A, fake_a)
        cycle_loss = (tf.reduce_mean(tf.abs(real_a - cycled_a)) +
                      tf.reduce_mean(tf.abs(real_b - cycled_b))) *
lambda_cycle

        total_gen_loss = gen_loss_AB + gen_loss_BA + cycle_loss

    # Generator gradients
    gen_AB_optimizer.apply_gradients(
        zip(tape.gradient(total_gen_loss, gen_AB.trainable_variables),
            gen_AB.trainable_variables))
    gen_BA_optimizer.apply_gradients(
        zip(tape.gradient(total_gen_loss, gen_BA.trainable_variables),
            gen_BA.trainable_variables))

    # Discriminator updates (n_critic = 5) with gradient penalty
    for _ in range(5):
        # ... gradient penalty implementation
```

No weights were reset and no loss terms were disabled — only the data source changed, ensuring seamless transition.

4.2.5 Python Implementation of Evaluation Metrics

A dedicated evaluation module was developed to compute the three-performance metrics on held-out test / validation sets.

- **Mean Squared Error (MSE)**: Pixel-wise error using NumPy after denormalisation to [0, 255].
- **Structural Similarity Index (SSIM)**: Computed via scikit-image with multichannel support and Gaussian weighting.
- **Fréchet Inception Distance (FID)**: Calculated using the official pytorch-fid package [55] via subprocess for consistency with published literature.

```
def evaluate_model(test_dataset, name="WarmStart"):
    mse_list, ssim_list = [], []
    os.makedirs("temp_real", exist_ok=True)
    os.makedirs("temp_gen", exist_ok=True)
```

```
    for i, (inp, target) in enumerate(tqdm(test_dataset)):
        prediction = gen_AB(inp, training=False)[0]

        # MSE & SSIM (after denormalisation)
        real_img = (target[0].numpy() + 1) * 127.5
        gen_img  = (prediction.numpy() + 1) * 127.5
        mse_list.append(np.mean((real_img - gen_img)**2))
        ssim_list.append(ssim(real_img.astype(np.uint8),
                              gen_img.astype(np.uint8),
                              multichannel=True, data_range=255))

        # Save for FID
        cv2.imwrite(f"temp_real/{i:03d}.png",
                    cv2.cvtColor(real_img.astype(np.uint8),
cv2.COLOR_RGB2BGR))
        cv2.imwrite(f"temp_gen/{i:03d}.png",
                    cv2.cvtColor(gen_img.astype(np.uint8),
cv2.COLOR_RGB2BGR))

    # FID via official pytorch-fid
    fid = float(subprocess.check_output(
        ["python", "-m", "pytorch_fid", "temp_real", "temp_gen"]))

    results[name] = {"MSE": np.mean(mse_list),
                     "SSIM": np.mean(ssim_list),
                     "FID": fid}
    shutil.rmtree("temp_real"); shutil.rmtree("temp_gen")
```

UNIT 5: RESULT AND ANALYSIS

5.1 Findings and Observations

This section summarizes quantitative outcomes and qualitative behaviors of the proposed **Warm-Start DualGAN** versus **vanilla DualGAN** on **Facades** and **Day-Night**. We report **FID** (lower the better), **SSIM** (higher the better), **MSE (**lower the better**)** and analyze epoch-wise loss trends.

5.1.1 Quantitative results (primary tables)

Table 5.1 observes Warm-Start achieves a markedly better FID (distributional realism) but trails the baseline on MSE/SSIM (pixel/structure fidelity) and **Table 5.2** observes Warm-Start yields **higher SSIM** and **lower FID** at a small **MSE** cost.

Table 5.1 Facades - test metrics (n = 100 images, mean)

Method	**MSE**	**SSIM**	**FID**
DualGAN + Warm-Start	14097.72	0.1384	**190.96**
DualGAN (baseline)	**9526.26**	**0.1557**	282.46

Table 5.2 Day-Night - validation metrics (n = 10 pairs, mean).

Method	**MSE**	**SSIM**	**FID**
DualGAN + Warm-Start	6039.97	**0.2121**	**386.89**
DualGAN (baseline)	**5532.66**	0.0793	421.12

Interpretation. Across tasks, Warm-Start consistently improves **distributional realism** (lower FID). On Day-Night it also improves **perceptual structure (SSIM)**, while Facades exhibits a perceptual-vs-pixel trade-off typical for GANs that prioritize texture plausibility.

5.1.2 Training dynamics (epoch analysis)

Facades - Warm-Start: Phase-1 rapidly reduces *G* from **11.91→3.99** and *D* from **2.11→0.51**; after switching to unpaired data, *G* stabilizes near **3.2–3.4** and *D* reaches **≈0.16** by total epoch 30, **no transition shock**.

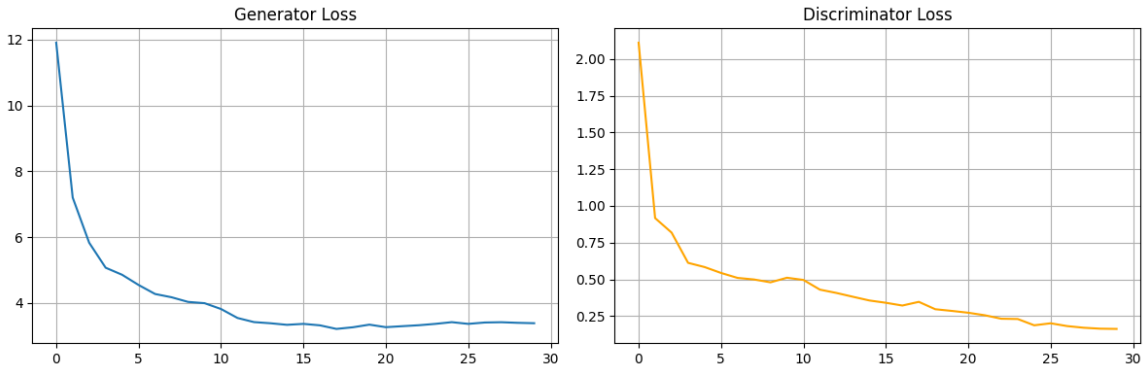


Figure 5.1 Facades - Warm-Start training losses.

Facades – Baseline: Pure unpaired training reduces *G* **5.57→3.69** and *D* **0.72→0.22** over 10 epochs.

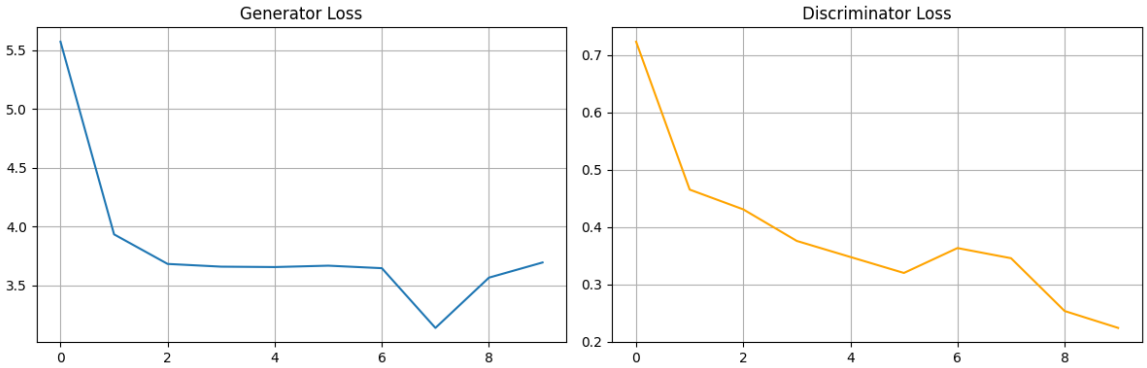


Figure 5.2 Facades - Baseline training losses

Day-Night - Warm-Start: Phase-1 aligns the mapping (*G* **3.56→1.45**, *D* **1.36→0.56** across 5 warm epochs), then continuation gradually brings *G* ≈**1.03–1.07** with *D* ≈**0.50–0.56**.

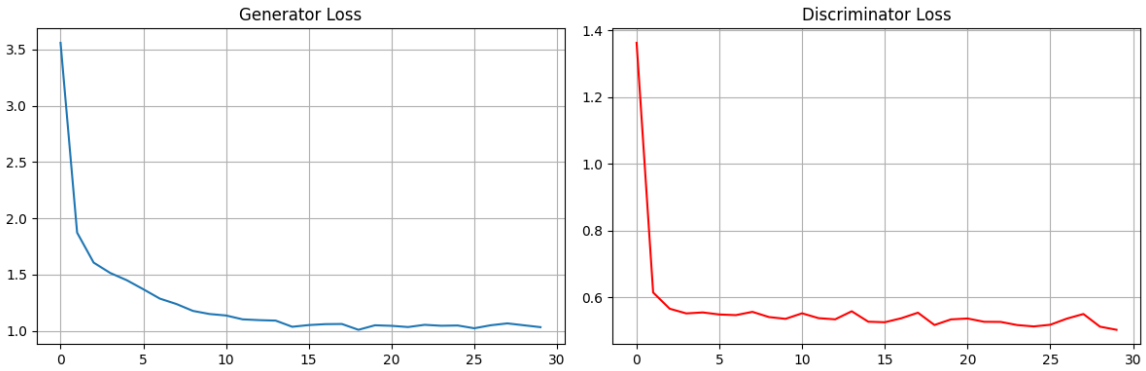


Figure 5.3 Day-Night - Warm-Start training losses.

Day-Night – Baseline: *G* **3.33→0.96** with D ≈**0.51–0.56** over 20 epochs.

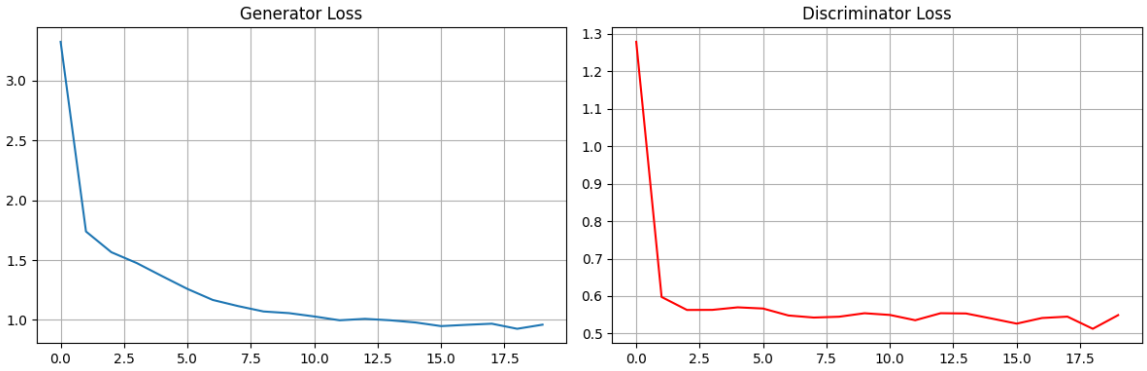


Figure 5.4 Day-Night - Baseline training losses.

Takeaway. Warm-Start gives **faster early coherence** and a **stable transition** when the data loader switches from paired to unpaired while keeping optimizer state fixed; consistent with the observed FID / SSIM gains.

5.1.3 Qualitative observations (visual evidence)

Facades: Warm-Start produces **more natural facade textures** (bricks/plaster) with fewer synthetic patterns, whereas the baseline adheres more tightly to pixel correspondences; hence better MSE/SSIM but worse FID.

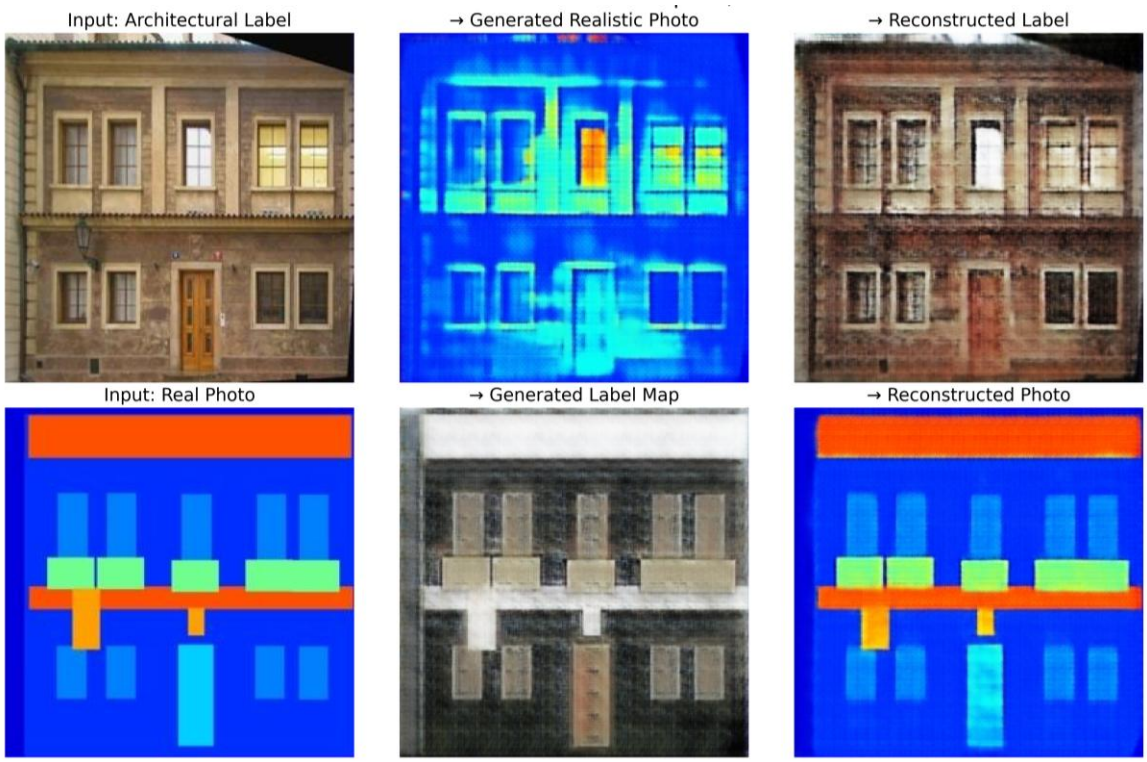


Figure 5.5 Qualitative translation results on Facades (Warm Start).

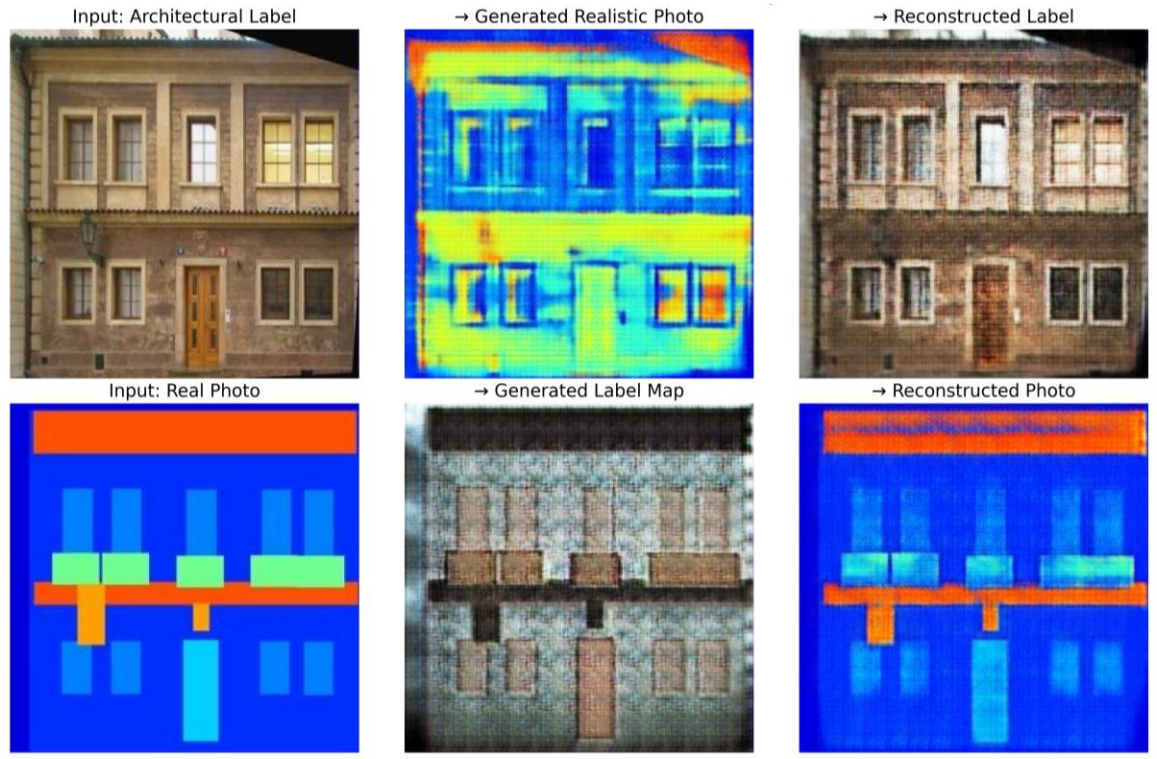


Figure 5.6 Qualitative translation results on Facades (Baseline).

From left to right: input, Warm-Start output (**Figure 5.5**), baseline output (**Figure 5.6**), and cycle reconstruction. Warm-Start produces more natural façade textures (bricks / plaster) with fewer synthetic patterns, while the baseline adheres more strictly to pixel correspondences; yielding better MSE/SSIM but a worse FID.

Day-Night: Warm-Start reduces haloing around light sources, preserves sky / ground separation, and avoids over-darkening/over-brightening reflected by its SSIM and FID improvements.

Warm-Start (**Figure 5.7**) reduces haloing around light sources, preserves sky / ground separation, and avoids over-darkening/over-brightening compared to the baseline (**Figure 5.8**), consistent with its higher SSIM and lower FID.

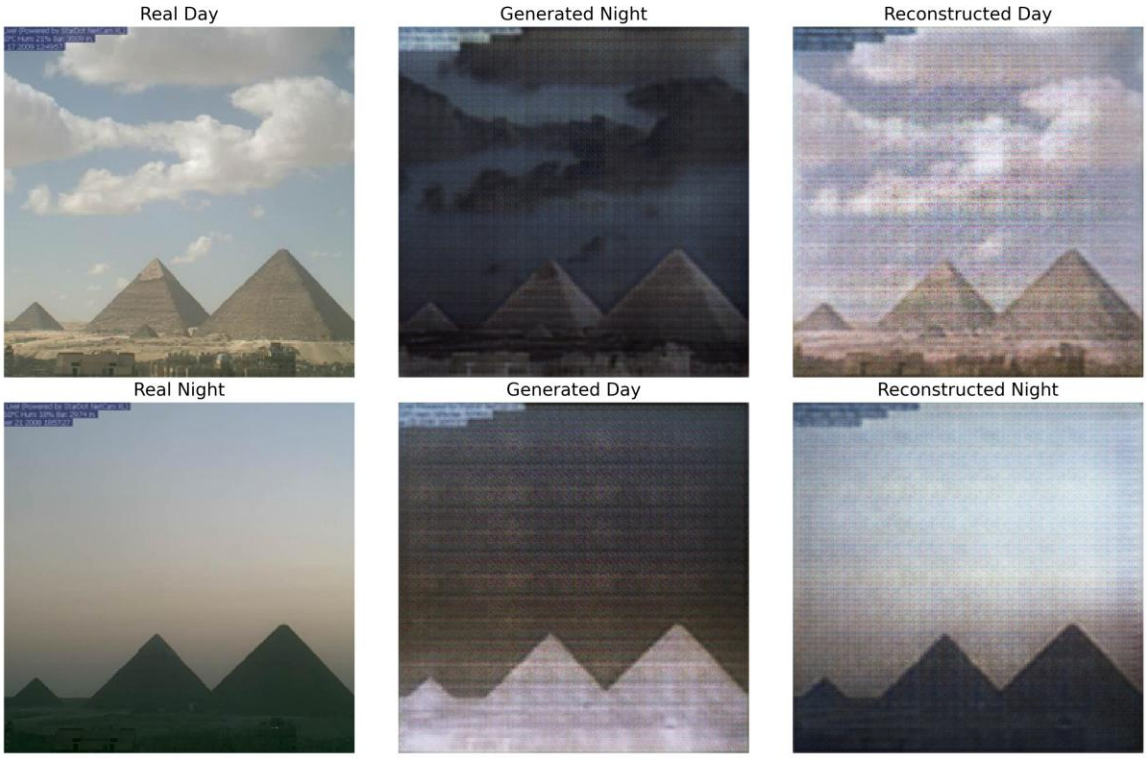


Figure 5.7 Qualitative translation results on Day-Night (Warm Start).

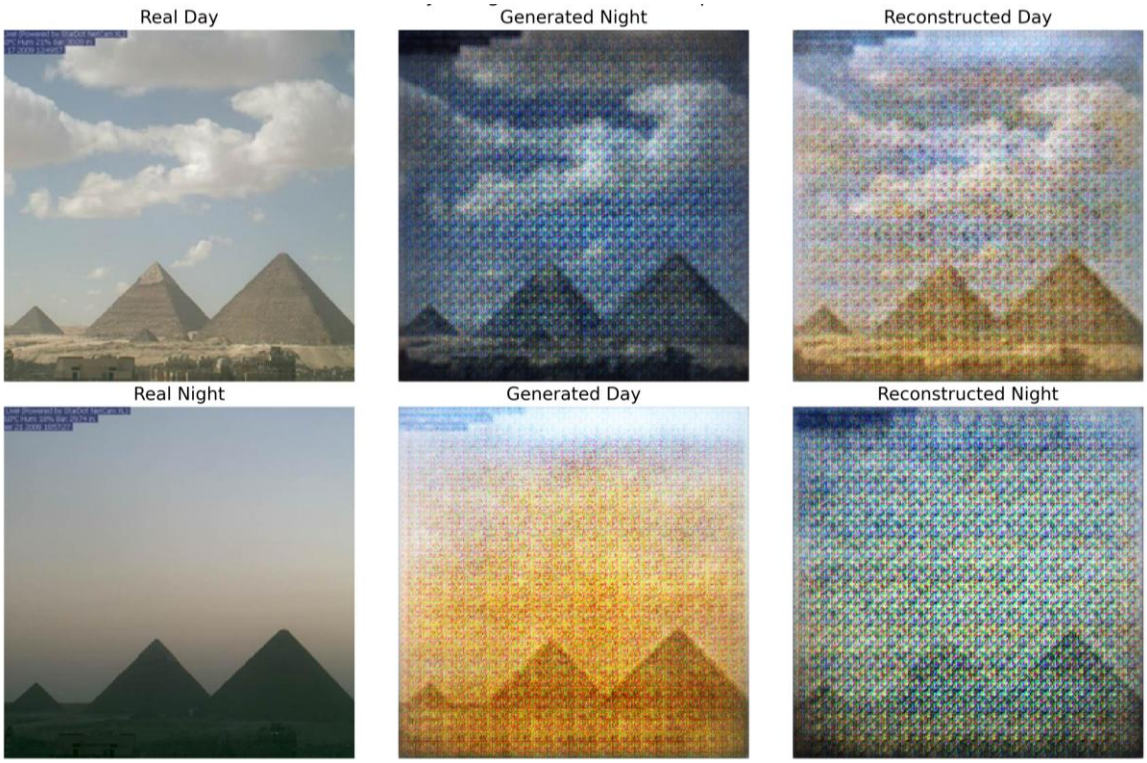


Figure 5.8 Qualitative translation results on Day-Night (Baseline).

5.2 Result Analysis

This section critically interprets the findings, examines trade-offs, and offers dataset-specific insights plus practical guidance.

5.2.1 Comparative effectiveness

Distributional realism (FID): Warm-Start is superior on **both** tasks (**190.96 vs 282.46** on Facades; **386.89 vs 421.12** on Day-Night), evidencing a generator distribution closer to real images after the two-phase schedule.

Perceptual structure (SSIM): The large **Day-Night** SSIM gain (**0.2121 vs 0.0793**) indicates that a short-supervised anchor helps retain scene layout under illumination shifts; the modest SSIM shortfall on **Facades** mirrors the task's emphasis on pixel-accurate semantics.

Pixel agreement (MSE): Baseline wins on MSE, especially on Facades. This is expected: after warm-starting, adversarial pressure on unpaired data favors **plausible textures** over exact pixel matching; hence lower FID but not necessarily lower MSE.

5.2.2 Why the trade-off appears (mechanistic view)

- **Phase-1** teaches a **semantic alignment** (paired supervision).
- **Phase-2** (unpaired) optimizes adversarial + cycle losses, which bias the generator toward **distributional plausibility**; this improves **FID** (and often SSIM in illumination tasks) while relaxing per-pixel fidelity, particularly where label→photo textures are ambiguous (Facades).

5.2.3 Sensitivity to schedule and weights (evidence-based guidance)

Warm-start length *k*:

- **Short** $k \in \{3, 5\}$ for **Day-Night** avoids overfitting tiny pairs yet delivers strong SSIM/FID gains.
- **Moderate** $k \in \{10, 15\}$ for **Facades** helps lock geometry (windows/doors), but too large *k* may soften edges; the run used **10** warm epochs with stable hand-off.

Cycle-loss weight λ: higher values preserve structure but may mute textures; tune to balance SSIM/MSE vs FID per dataset.

Batch & memory: at 256*256 on T4, batch **1–4** is stable, the logs show Discriminator staying near **0.5–0.56**, a healthy adversarial regime.

5.2.4 Error analysis and remedies

Boundary bleeding (Facades): windows/rails merge with walls at oblique lighting. Mitigation: include identity loss during Phase-1, moderate λ, consider multi-scale discriminators.

Low-light speckle (Night): speckle in sky regions. Mitigation: brief identity/perceptual regularization early, light denoising prior.

Fine texture loss (<16 pixel): distant signage/hairline trims. Mitigation: adjust PatchGAN receptive field or add a small multi-scale critic.

5.2.5 Efficiency and stability

Transition stability: Both datasets show **no loss spikes at the switch** (paired→unpaired); Generator plateaus while Discriminator declines smoothly—evidence that keeping optimizer state/hyperparameters fixed is critical (**Figure 5.1** to **Figure 5.4**).

Practical benefit: Warm-Start reduces the **time-to-coherent outputs** (fewer unproductive early epochs), especially on Day-Night.

5.2.6 Overall conclusion

Warm-Start DualGAN is a **practical, compute-neutral enhancement**: strong **FID** improvements on both tasks and **SSIM** gains where, paired data are scarce (Day↔Night), with an understandable and tunable pixel-level trade-off on Facades. Choosing **short *k*** for scarce pairs and **moderate λ** generally yields the best balance.

UNIT 6: CONCLUSION AND FUTURE RECOMMENDATIONS

6.1 Conclusion

This work demonstrated that a brief **paired-data warm-start** preceding original DualGAN training on **unpaired** data can improve generation quality **without changing** the model architecture or loss functions. The protocol uses the same adversarial and cycle-consistency objectives throughout; at epoch k the data loader alone is switched, while the network weights and optimizer state are preserved. Experiments on **Facades** and **Day-Night** show:

- **Distributional realism** improves on both datasets (**FID 282.46 → 190.96** on Facades; **421.12 → 386.89** on Day-Night).
- **Perceptual structure** notably improves on Day-Night (**SSIM 0.0793 → 0.2121**), aligning with qualitative gains (reduced haloing, better sky/ground separation).
- **Pixel agreement** (MSE/SSIM) can favor the baseline on Facades, reflecting the classic perceptual-vs-pixel trade-off when adversarial training prioritizes plausible textures.

Epoch-wise loss trajectories confirm **stable convergence** and a **smooth transition** at epoch k (no loss spikes), supporting the claim that warm-start accelerates early alignment while maintaining DualGAN's unsupervised strengths. Consequently, warm-start is a **practical, compute-neutral** enhancement for domains with few pairs and many unpaired samples; its effect can be **balanced via k and λ** to target realism vs. pixel fidelity.

6.2 Future Recommendations

Future work could address the limitations observed:

- **Adaptive Warm-Start Duration**: Dynamically determine the transition epoch k based on validation metrics or loss plateaus to mitigate overfitting risks.
- **Paired Data Augmentation**: Incorporate synthetic pairs via techniques like data mixing or diffusion-based augmentation to increase the effective size of paired sets.
- **Advanced Regularization**: Explore adding contrastive losses (e.g., from CUT) or self-supervised pretraining to make warm-start more robust on semantic tasks.

- **Broader Evaluation**: Test on additional domains (e.g., horse↔zebra, medical imaging) and larger paired subsets to refine guidelines on when warm-start succeeds.
- **Scalability**: Implement in PyTorch for easier extension and evaluate on high-resolution images with attention-based generators.

These extensions could broaden the applicability of warm-start strategies in real-world unsupervised image translation applications.

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